Tilak Patel

315-720-8202 | tilakny@gmail.com | LinkedIn | GitHub | tilakpatell.com | US Citizen | New York | Boston |
Availability: June 2025 - January 2026

EDUCATION

Northeastern University **May 2027**
Khoury College of Computer Sciences
Major: Candidate for B.S. Computer Science (AI)
Minors: Computer Engineering and Robotics
GPA: 3.54/4.0 | Dean's List: Fall 2024
Relevant Coursework: Computer Systems, Network Fundamentals, LLM-Integrated Systems, Object-Oriented Design, Algorithms, Robotic Science & Systems, Embedded Design: Enabling Robotics

TECHNICAL SKILLS

Languages & Databases: Python, Java, C/C++, HTML/CSS, Assembly (x86), JavaScript/TypeScript, ROS2, MongoDB
AI/ML & Computer Vision: PyTorch, NumPy, Pandas, Scikit-learn, LLMs, NLP, RAG, Multi-Agent Systems, Hugging Face Transformers, OpenCV, Langchain
Systems & Hardware: Linux, Operating Systems, Virtual Machines, Embedded Systems, DE1-SoC Board, FPGA Design, Memory Mapping, PWM, GPIO, Jetson Orin Nano, Raspberry Pi 5
Development Tools: Git, Docker, AWS (EC2), GDB, Valgrind, Pytest, JUnit, Jupyter, Postman, VS Code, Gradle, Maven
Frameworks & Frontend: Node.js, FastAPI, Flask, React, Material UI, Tailwind CSS

EXPERIENCE

Northeastern High Performance Computing Lab **Jan 2025 - Present**
Researcher Boston, MA
- Optimizing MPI checkpointing systems for large-scale supercomputing applications through MANA/DMTCP framework, focusing on improving communication and synchronization between processes
- Conducting performance analysis on Northeastern's Discovery cluster using flame graphs to identify and reduce checkpoint/restart overhead in HPC workloads
- Utilizing Discovery's advanced computing infrastructure (50,000+ CPU cores) at the Massachusetts Green HPC Center for distributed computing research

Later **Sept 2024 – Dec 2024**
R&D Engineer Boston, MA
- Architected vision-language system (CLIP, BLIP, Llama), processing 1M+ images with speed and modularity.
- Engineered optimized CLIP-BLIP pipeline combining embedding generation and image captioning, achieving dynamic labeling and improved performance.
- Developed FAISS indexing with KMeans clustering enabling similarity search across a million-scale image dataset.

Empowerreg AI **July 2024 – Sept 2024**
Generative AI Intern Seattle, WA
- Engineered production-ready visualizations for Risk Annotation Matrix Product, enhancing data interpretation.
- Developed visualizations for multi-dimensional risk and complaint data analysis of medical devices from LLMs.
- Optimized processing of 1,000+ lines of JSON medical device data using FastAPI, reducing transmission time.

PROJECTS

FUSE File System | C, FUSE, Linux, Memory Mapping, inode Architecture **Dec 2024**
- Engineered inode design for metadata, direct/indirect blocks, and 4KB memory mapping.
- Developed low-level implementations, including directory manipulation, block allocation, and bitmap management.
- Built essential file operations (read, write, create, delete) with proper error handling and memory safety checks.

Finance Platform | React, TypeScript, Python FastAPI, MongoDB, TailwindCSS, Framer Motion **Dec 2024**
- Engineered a full-stack financial platform with real-time stock data and portfolio analytics using Polygon.io API.
- Implemented secure user authentication and advanced trading features with MongoDB and FastAPI backend.
- Developed an AI financial advisor using Claude API for personalized market analysis and portfolio recommendations.

Interactive AI Tutor | Python, LangChain, Gradio, Tavily API, Llama 3, OpenAI API **Dec 2024**
- Designed an intelligent tutoring system that provides guided problem-solving support through hints.
- Built modular agents for query analysis, multi-level hint generation, and contextual learning using LangChain.
- Integrated Tavily API for real-time search capabilities to enhance response relevance and user learning.